

Econ 101A

Section 4

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1 Envelope Theorem

1.1 The Theorem

Suppose $x^*(p)$ solves the following optimization problem:

$$\max_x f(x; p)$$

In this example:

- p is **exogenous**, meaning that it is determined independently of other variables.
- $x^*(p)$ is **endogenous**, as it is a function of other (exogenous) variables p .

Define $F(p) = f(x^*(p), p)$ (this expression is now just a function of p , since at the optimum, x is endogenously determined by p). Then, we have:

$$\begin{aligned} \frac{dF(p)}{dp} &= \underbrace{\frac{\partial f(x; p)}{\partial x} \bigg|_{x=x^*}}_{\text{By FOC, this is 0}} \times \frac{\partial x^*}{\partial p} + \frac{\partial f(x; p)}{\partial p} \bigg|_{x=x^*} \\ &= \frac{\partial f(x; p)}{\partial p} \bigg|_{x=x^*} \end{aligned}$$

1.2 What does this mean?

It's very easy to calculate the derivatives at the maximum of an objective function! The total differential of the function $F(p)$ with respect to p equals the partial derivative with respect to p at $x = x^*$ (this means that you can disregard the indirect effects!).

- When differentiating a function with respect to an exogenous variable, typically you do the following:
 1. Write the function in terms of your exogenous variable only (in the above notation, this means that you would substitute x with its expression in terms of p , so $x^*(p)$).
 2. Then, take the derivative with respect to the exogenous variable (i.e. p)
- Using the Envelope Theorem, you can calculate the derivative at the optimum as follows:

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1. Differentiate the objective function with respect to the desired exogenous variable (holding all other variables constant, including the endogenous variable that maximizes your objective)
2. Now you have an expression (the derivative) written in terms of your endogenous variables. Plug in their optimal values of the endogenous variables (i.e. $x^*(p)$), and you're done!

2 Constrained Maximization

Suppose that you want to maximize $f(x; p)$ such that $h(x; p) = 0$, and x is an n -dimensional vector.¹ The steps:

1. Write the Lagrangian function:

$$\mathcal{L}(\lambda, x; p) = f(x; p) - \lambda h(x; p).$$

Note that λ is joining your x 's as an additional endogenous variable.

2. Your (necessary) first order conditions (setting $\nabla \mathcal{L}(\lambda^*, x^*; p)$ will now give your $n + 1$ equations in $n + 1$ unknowns. To get your candidate solutions, take the first derivatives of $\mathcal{L}$ with respect to all x_i 's and λ and set all $n + 1$ of these equations equal to zero. You will notice that you will get the following:

$$\begin{aligned} h(x^*; p) &= 0 \\ \frac{\partial f(x^*; p)}{\partial x_i} &= \lambda^* \times \frac{\partial h(x^*; p)}{\partial x_i} \quad \text{for all } i \end{aligned}$$

3. To check the sufficient second order condition for a constrained maximum (or minimum), we use the **bordered Hessian**, which is calculated simply as the Hessian of $\mathcal{L}(\lambda, x; p)$. However, we need a different condition than before to establish whether our candidate solution is a max or min.
4. In the $n = 2$ (two x 's) and one constraint case, the sufficient conditions are as follows:
 - the determinant of the bordered Hessian is *positive* for a constrained *maximum*
 - the determinant of the bordered Hessian is *negative* for a constrained *minimum*
5. Also, note that when calculating the derivative with respect to your exogenous variable p , the Envelope Theorem applies to the partial derivative of the *Lagrangian*, NOT the partial in the objective function. In other words, the Envelope Theorem for constrained optimization is as follows:

$$\frac{df(x^*(p); p)}{dp} = \frac{\partial \mathcal{L}(\lambda^*(p), x^*(p); p)}{\partial p}$$

3 (A Little Bit of) Multivariate Calculus

Let $i = 1, 2, \dots, m$ denote the rows of a matrix (or the elements of a column vector) and $j = 1, 2, \dots, n$ the columns. (Note that for a square matrix $m = n$.)

3.1 Scalar multivariate functions:

- Here, we have one variable on the left-hand side (LHS) and many variables on the right-hand side (RHS).

$$y = f(x_1, x_2, \dots, x_m)$$

¹Here, we will assume that h is scalar-valued, meaning that we only have one constraint. In lecture, we were more general, and allowed ourselves to have more than one constraint.

- y is a "scalar" meaning it is just one number, not a vector. (You can also think of y as a 1×1 matrix.)
- The x'_i s on the RHS make up a vector. (We will often, but not always, use **bold** for vectors.)

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}$$

- Then we can write the function as a function of a vector: $y = f(x_1, x_2, \dots, x_m)$ is the same as $y = f(\mathbf{x})$.

3.2 First and second derivatives of scalar multivariate functions:

3.2.1 Gradient:

- The first derivative of a scalar multivariate function, f , is called the "gradient" of f , and is notated ∇f .
- The gradient is the vector of partial derivatives of the function with respect to each of the RHS variables. Thus:

$$\nabla f = \left[\frac{\partial f}{\partial x_i} \right] = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_m} \end{bmatrix}$$

- **Problem #1:**² Derive the gradient of $y(x_1, x_2; a)$ over $(x_1, x_2) \in \mathbb{R}^2$, where

$$y(x_1, x_2; a) = \ln(x_1) + e^{ax_2}$$

3.2.2 Hessian:

- The second derivative of a scalar multivariate function, f , is called the "Hessian" of f , and is notated H_f .
- The Hessian is the matrix of all of the second- and cross-partial derivatives of the function. Thus:

$$H_f = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right] = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_m} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_m \partial x_1} & \dots & \dots & \frac{\partial^2 f}{\partial x_m^2} \end{bmatrix}$$

- To form this matrix, notice that the j th column is the gradient differentiated with respect to the j th input variable.
- Notice that the Hessian is square and, provided f is "smooth" enough, symmetric (by Clairaut's Theorem, which states that $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$ whenever $\frac{\partial^2 f}{\partial x_i \partial x_j}$ and $\frac{\partial^2 f}{\partial x_j \partial x_i}$ are continuous).
- **Problem #2:** Derive the Hessian of $y(x_1, x_2; a)$ over $(x_1, x_2) \in \mathbb{R}^2$, where

$$y(x_1, x_2; a) = -(ax_1 - 1)^2 - (x_2 - 2)^2$$

²Note that the semicolon ; is used to distinguish between *variables* and *parameters*. Everything before the semicolon indicates it is a variable, meaning that this quantity is subject to change, and you can change its value, take derivatives of it, etc. Everything *after* the semicolon is a *parameter*, which is a fixed quantity and you treat it as a constant.

4 Exercises

4.1 Envelope Theorem

A student wants to maximize her grade (g) on an essay. She selects her time spent writing (t) in order to achieve her top grade. Her time spent writing is a function of the noise at her local library (q), which is out of her control (or *exogenous*):

$$g = -(2t - q)^2 - t^2 + 100$$

In short, she is solving the following problem: $\max_t g(t; q)$. The F.O.C. for this problem is given by: $-4(2t - q) - 2t = 0$ (check this). From our FOCs, we can solve for our optimal t in terms of q , finding that $t^*(q) = \frac{2q}{5}$. Using this information, calculate the derivative of g with respect to q at the optimal level of t (i.e., the rate at which her grade changes in response to a change in library “quality”) in two different ways:

1. Plug in $t^*(q) = \frac{2q}{5}$ for t into our original expression for g , and then differentiate with respect to q .
2. (Envelope Theorem) Calculate the partial derivative $\frac{\partial g}{\partial q}$, and evaluate when $t = t^*(q)$.
3. How do these derivatives compare?

4.2 Constrained Optimization

Each day, Elisabetta has quasi-linear utility over leisure l and consumption c . In particular, she maximizes $U(l, c; w) = \phi(l) + c + \psi(w)$. She can only spend her endowment M and wages earned wh , where w is the hourly wage rate and $h = 24 - l$ are her hours spent working. The price of consumption is p .

1. Derive Elisabetta’s budget constraint.
2. Write down the maximization problem of the worker with respect to c and l with the relevant constraints. (Assume that the budget constraint is satisfied with equality.)
3. Write down the Lagrangian, and derive the first order conditions with respect to c , l , and λ .
4. Simplify the problem to a system of two equations and two unknowns c^* and l^* .
5. Assume that $\phi'(l^*) > 0$ and $\phi''(l^*) < 0$ [Utility increases with more leisure but at a decreasing rate]. What would happen with the optimal leisure l^* if w increases? Provide intuition. If time permits, use IFT to formally find: $\frac{\partial l^*}{\partial w}$.